Board Sheet — Lesson 10: Analysis of Residuals

MA153X Mathematical Modeling

Name: ______________________________ **Date:** ____________________

What is a Residual? (Reading 2.3)

Definition: $e_i =$ ______________ $-$ ______________

If the residual is...	Then the model...
Positive ($e_i > 0$)	
Negative ($e_i < 0$)	
Zero ($e_i = 0$)	

Computing Residuals

Using your model from Lessons 8–9, compute residuals for a sample of training soldiers:

Soldier	Height (in)	Actual Weight (lbs)	Fitted ($\hat{y}$)	Residual (e)	Interpretation
S002	74	239			
S003	63	147			
S004	70	169			
S005	68	159			
S006	72	211			
S007	73	239			
S008	71	202			
S009	64	136			
S010	67	194			
S011	74	208			

Excel: Fitted = `= intercept_cell + slope_cell * Height_cell`; Residual = `= Actual - Fitted`

The Residual Plot

Create a scatter plot: **Fitted Values** ($\hat{y}$) on x-axis, **Residuals** (e) on y-axis. Add a line at $e = 0$.

A good residual plot shows:

1. ____
2. ____
3. ____

Three Bad Patterns

Curved Pattern — Which assumption violated? ____________________

Fan Shape (heteroscedasticity) — Which assumption violated? ____________________

Outliers — What should you do? ____

____

Distribution of Residuals

Mean of residuals is always ________ **(why?)** ____

Standard Deviation of Residuals (s_e)

$$s_e = \sqrt{\frac{\sum_{i=1}^{n} e_i^2}{n-2}}$$

Why $n-2$ instead of $n-1$? ____

Compute s_e for our model ($n = 37$):

$s_e =$ __________ lbs. **Interpretation:** ____

Informal Normality Check

- Are most residuals small (close to zero)? ________
- Is the distribution roughly symmetric? ________
- Any extreme outliers? ________

____

Residual Analysis Checklist (from Reading 2.3)

Step	Check	Result
1	Compute $\hat{y}_i$ and e_i for all training observations	Done above
2	Mean of residuals ≈ 0?	
3	Residual plot: random scatter?	[ ]
4	No curved pattern?	[ ]
5	No fan shape?	[ ]
6	No extreme outliers?	[ ]
7	s_e acceptable for the decision?	$s_e =$ _____ lbs

Connecting to the Four Assumptions

Assumption	What Residual Analysis Shows
Linearity	
Independence	
Constant variance	
Normality	

Final Evaluation

Based on ALL diagnostic evidence (Lessons 8–10), is a linear regression model appropriate for predicting weight from height?

Takeaways

Why can't we just use R^2 to evaluate a model?

What is the single most important diagnostic for linear regression?
